

Daniel C. Patiño

Senior Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 6+ years shipping production models in retail and fintech. Primary platform Databricks: ML governance, MLflow, Delta Lake, workflows, and deployment. Python, SQL, Spark, TensorFlow, and PyTorch; strong data foundations (BigQuery, ETL) and GANs research.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Mar 2023 - Present
Farmatodo Bogotá, Colombia

- Defined team-wide ML governance guidelines and best practices on Databricks (experimentation, deployment, and traceability)
- Lead production ML on the Databricks lakehouse (Delta Lake, workflows, monitoring), integrated with the enterprise data warehouse and analytical domains
- Co-led the BigQuery Data Warehouse (5+ sources: Oracle, Braze, Amplitude, Algolia) and evolution toward Databricks with Terraform IaC
- Architected ETL/ELT pipelines in Python and Spark with proactive monitoring, alerts, and error recovery
- Built user affinity and multi-channel marketing attribution at retail scale with impact and ROI dashboards and analysis

Data Engineer Apr 2022 - Mar 2023
Ultracom (Cliente: Bancolombia) Medellín, Colombia

- Maintained and evolved components of Bancolombia's enterprise Data Warehouse (SQL/Python, batch jobs, dimensional adjustments and ingestion for business needs)
- Migrated Python pipelines to Cloudera CDP Public Cloud, aligning dependencies and deployment for more elastic scaling under load
- Deployed Django APIs on AWS (DynamoDB, Glue, S3, CloudFormation) exposing lake data and operations to internal consumers
- Applied code reviews and production deployment quality gates (testing standards) for Python services on AWS

Data Scientist & ML Engineer Jul 2018 - Mar 2022
Frontierx Medellín, Colombia

- Researched and developed generative models (GANs) for synthetic commodities data, resulting in indexed scientific publication
- Implemented and deployed ML-based algorithmic trading strategies across multi-commodity markets (metals, energy, agri), working with trading desks
- Built predictive models for portfolio positioning in the Spanish energy market
- Processed high-resolution satellite imagery with Robosat (computer vision) to enrich territorial signals for predictive models

EDUCATION

Master's Degree in Engineering May 2019 - Nov 2022
Universidad de Antioquia Medellín, Colombia

- Research focus: Deep Learning under Professor Raúl Ramos Pollán's supervision
- Thesis: Impact of synthetic data (GANs) on commodities trading predictions
- Indexed scientific publication in Research in International Business and Finance (2022)

Systems Engineering Sep 2013 - Aug 2018
Universidad Industrial de Santander Bucaramanga, Colombia

- Thesis: Urban socioeconomic stratification estimation using CNNs applied to satellite imagery

PUBLICATIONS

Synthetic data generation with deep generative models to enhance predictive tasks in trading strategies
Research in International Business and Finance, 2022

TECHNICAL SKILLS

Languages:
Python, SQL, JavaScript/TypeScript

Databricks & Lakehouse:
Databricks (Lakehouse), Apache Spark / PySpark, Delta Lake, Workflows & Jobs, Unity Catalog, MLflow, Terraform (Data-bricks)

Machine Learning & AI:
TensorFlow, PyTorch, scikit-learn, GANs, CNNs, Time Series, MLOps / Model governance

Data Analysis:
Pandas, NumPy, Matplotlib, Jupyter, Feature Engineering

Data Engineering:
ETL/ELT Pipelines, Data Warehousing, Data Modeling, Data Quality

LANGUAGES

Spanish	Native
English	C1 Advanced

CERTIFICATIONS

TensorFlow Developer Professional Certificate
Coursera (DeepLearning.AI) • Sep 2019

Deep Learning Specialization
Coursera (DeepLearning.AI) • Jun 2019

Build Basic Generative Adversarial Networks (GANs)
Coursera • May 2021

EF SET English Certificate (C1 Advanced)
EF Education First • Feb 2023